DESIGN DATA BOOK — SHAFT COUPLINGS

Standard Mechanical Engineering Design Data, Formulas & Solved Protocols (IS: 2048 - 1962 / IS: 4218 - 1976)

Shaft Couplings Classification & Design Index

Coupling Type	Design Requirements & Conditions	Data Book Section
Muff (Sleeve) Coupling	Design shaft d, cast iron sleeve D, L, and key w, t, l	Section 1
Clamp / Split-Muff Coupling	Design shaft d, sleeve D, L, key, and clamping bolts d_b	Section 2
CI Protective Flange Coupling	Design hub, key, web t_f, bolts n, d_1, and flange D_2, t_p	Section 3
Protective Flange Coupling	Design for peak torque $T_{\max} = 1.25 T_{\text{mean}}$ and bolt sizing	Section 4
Flange Coupling (Rigidity Limit)	Size shaft d for strength & rigidity ($\theta \leq 1^{\text{deg}}$ in $20d$)	Section 5
Rigid Flange Coupling	Direct torque $T = 250\ \text{N} \cdot \text{m}$ design for 4 bolts	Section 6
Flanged Coupling Analysis	Analyze fixed shaft $d = 35\ \text{mm}$ for key length l & power P	Section 7
Marine Engine Forged Flange	Design $3\ \text{MW}$ engine shaft d, 8 bolts ($D_1 = 1.6d$), and t_f	Section 8
Bushed-Pin Flexible Coupling	Design flexible pins, rubber bushes, hub, key & web t_f	Section 9

SECTION REF 1: Table 13.1 — Standard Key Proportions (IS: 2048 - 1962)

Shaft Dia d (mm)	Key Width w (mm)	Key Thick t (mm)	Shaft Dia d (mm)	Key Width w (mm)	Key Thick t (mm)
Up to 6	2	2	Up to 85	25	14
Up to 8	3	3	Up to 95	28	16
Up to 10	4	4	Up to 110	32	18
Up to 12	5	5	Up to 130	36	20
Up to 17	6	6	Up to 150	40	22
Up to 22	8	7	Up to 170	45	25
Up to 30	10	8	Up to 200	50	28
Up to 38	12	8	Up to 230	56	32
Up to 44	14	9	Up to 260	63	32
Up to 50	16	10	Up to 290	70	36
Up to 58	18	11	Up to 330	80	40
Up to 65	20	12	Up to 380	90	45
Up to 75	22	14	Up to 440	100	50

Empirical Key Design Rules & Strength Constants:

1. Rectangular Sunk Key: Width $w \approx \frac{d}{4}$, Thickness $t \approx \frac{d}{6}$
2. Square Sunk Key: Width $w = t \approx \frac{d}{4}$ (used when allowable crushing stress $\sigma_c = 2\tau$)
3. Keyway Shaft Strength Factor: $e = 1 - 0.2\left(\frac{w}{d}\right) - 1.1\left(\frac{h}{d}\right)$

SECTION REF 2: Table 11.1 — Standard Metric Screw Threads (Coarse Series)

Designation	Pitch p	Nominal d	Pitch d_p	Core d_c	Stress Area A_c
M 6	1.0 mm	6.0 mm	5.350 mm	4.773 mm	20.1 mm²
M 8	1.25 mm	8.0 mm	7.188 mm	6.466 mm	36.6 mm²
M 10	1.5 mm	10.0 mm	9.026 mm	8.160 mm	58.3 mm²
M 12	1.75 mm	12.0 mm	10.863 mm	9.858 mm	84.0 mm²
M 16	2.0 mm	16.0 mm	14.701 mm	13.546 mm	157 mm²
M 18	2.5 mm	18.0 mm	16.376 mm	14.933 mm	192 mm²
M 20	2.5 mm	20.0 mm	18.376 mm	16.933 mm	245 mm²
M 24	3.0 mm	24.0 mm	22.051 mm	20.320 mm	353 mm²
M 27	3.0 mm	27.0 mm	25.051 mm	23.320 mm	459 mm²
M 30	3.5 mm	30.0 mm	27.727 mm	25.706 mm	561 mm²

SECTION 1: Solid Sleeve / Muff Coupling Design

System Parameters

Transmitted Power: $P = 40\ \mathrm{kW} = 40000\ \mathrm{W}$

Rotational Speed: $N = 350\ \mathrm{rpm}$

Shaft & Key Stresses: Shear $\tau_s = 40\ \mathrm{MPa}$, Crushing $\sigma_c = 80\ \mathrm{MPa}$ | Muff CI Shear: $\tau_c = 15\ \mathrm{MPa}$

Design Protocol

Part 1: Calculate design torque T and solve shaft diameter d

Part 2: Determine muff proportions D, L and verify induced CI shear stress τ_c

Part 3: Select key size from Table 13.1 and verify shearing τ_s and crushing σ_{cs}

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1. Transmitted Design Torque (T)

Torsional moment transmitted by connected shafts

$$T = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 40000}{2\pi \times 350}$$

$$= 1091.35\ \mathrm{N \cdot m}$$

$$= 1100 \times 10^3\ \mathrm{N \cdot mm}$$

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2. Shaft Diameter Sizing (d)

Torsional shear strength formula for solid shaft

$$d = \sqrt[3]{\frac{16T}{\pi\tau_s}}$$

$$= \sqrt[3]{\frac{16 \times 1.09135 \times 10^6}{\pi \times 40}}$$

$$= 51.8 \text{ mm}$$

3. Standard Shaft Diameter Selection

Select standard nominal shaft diameter

Adopt Standard Shaft Diameter $d = 55$ mm

4. Cast Iron Muff Dimensions (D, L)

Empirical sleeve proportions

$$D = 2d + 13 \text{ mm}$$

$$= 2(55) + 13$$

$$= 123 \text{ mm}$$

$$\approx 125 \text{ mm}$$

$$L = 3.5d$$

$$= 3.5 \times 55$$

$$= 192.5 \text{ mm}$$

$$\approx 195 \text{ mm}$$

5. Induced Shear Stress in Muff (τ_c)

Hollow shaft torsional shear check

$$\tau_c = \frac{16TD}{\pi(D^4 - d^4)}$$

$$= \frac{16 \times 1.09135 \times 10^6 \times 125}{\pi(125^4 - 55^4)}$$

$$= 2.96 \text{ MPa}$$

$$\leq 15 \text{ MPa} \quad (\textbf{SAFE \& SATISFACTORY})$$

6. Key Proportions Selection (Table 13.1)

Square key selected as $\sigma_c = 2\tau_s$

$$w = 18 \text{ mm}$$

$$t = 18 \text{ mm} \quad \text{(Square Key)}$$

$$l = \frac{L}{2} = \frac{195}{2} = 97.5 \text{ mm}$$

7. Induced Shear Stress in Key (τ_s)

Key shear strength verification

$$T = l \cdot w \cdot \tau_s \cdot \left(\frac{d}{2}\right)$$

$$1100 \times 10^3 = 97.5 \times 18 \times \tau_s \times \left(\frac{55}{2}\right)$$

$$\tau_s = \mathbf{22.8 N/mm^2}$$

$$\leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

8. Induced Crushing Stress in Key (σ_{cs})

Key crushing strength verification

$$T = l \cdot \left(\frac{t}{2}\right) \cdot \sigma_{\mathrm{cs}} \cdot \left(\frac{d}{2}\right)$$

$$1100 \times 10^3 = 97.5 \times \left(\frac{18}{2}\right) \times \sigma_{\mathrm{cs}} \times \left(\frac{55}{2}\right)$$

$$\sigma_{\mathrm{cs}} = \mathbf{45.6 N/mm^2}$$

$$\leq 80\ \mathrm{MPa} \quad (\mathbf{SAFE})$$

9. Design Output

Final Design: Shaft $d = 55$ mm, Muff $D = 125$ mm, $L = 195$ mm, Square Key $18 \times 18 \times 97.5$ mm

SECTION 1: Muff Coupling — MECHANICAL DIAGRAM

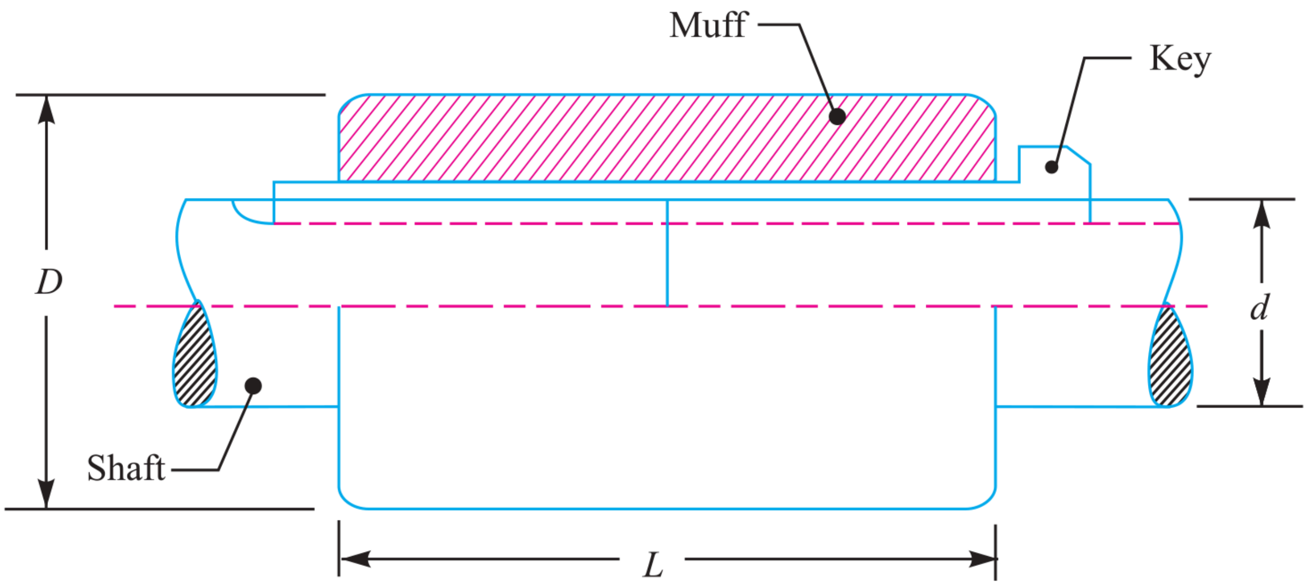

Fig. 13.10. Sleeve or muff coupling.

Figure 13.1: Sleeve or Muff Coupling Assembly & Empirical Proportions

SECTION 2: Clamp / Split-Muff Coupling Design

System Parameters

Power: $P = 30 \text{ kW} = 30000 \text{ W}$

Speed: $N = 100 \text{ rpm}$

Shaft & Key Shear: $\tau = 40 \text{ MPa}$

Bolt Tensile Stress: $\sigma_{\text{tb}} = 70 \text{ MPa}$ | Bolt Count: $n = 6$ | Friction Coefficient: $\mu = 0.3$

Design Protocol

Part 1: Calculate torque T and solve shaft diameter d

Part 2: Determine split muff dimensions D, L

Part 3: Select key dimensions from Table 13.1 and verify shear stress τ_k

Part 4: Calculate required bolt core diameter d_b from friction torque equation

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1. Transmitted Design Torque (T)

Torsional moment transmitted through clamp coupling

$$T = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 30000}{2\pi \times 100}$$

$$= 2864.8 \text{ N} \cdot \text{m}$$

$$= 2865 \times 10^3 \text{ N} \cdot \text{mm}$$

2. Shaft Diameter Sizing (d)

Torsional shear strength evaluation

$$d = \sqrt[3]{\frac{16T}{\pi \tau_s}}$$

$$= \sqrt[3]{\frac{16 \times 2.8648 \times 10^6}{\pi \times 40}}$$

$$= 71.45 \text{ mm}$$

3. Standard Shaft Diameter Selection

Select standard nominal shaft diameter

Adopt Standard Shaft Diameter $d = 75$ mm

4. Split Muff Dimensions (D, L)

Empirical muff proportions

$$D = 2d + 13 \text{ mm}$$

$$= 2(75) + 13$$

$$= 163 \text{ mm}$$

$$\approx 165 \text{ mm}$$

$$L = 3.5d$$

$$= 3.5 \times 75$$

$$= 262.5 \text{ mm}$$

$$\approx 265 \text{ mm}$$

5. Key Dimensions (Table 13.1) & Shear Check (τ_k)

Key proportions for $d = 75$ mm

$$w = 22 \text{ mm}, \quad t = 14 \text{ mm}, \quad l = L = 265 \text{ mm}$$

$$\tau_k = \frac{2T}{w \cdot d \cdot L}$$

$$= \frac{2 \times 2.8648 \times 10^6}{22 \times 75 \times 265}$$

$$= 13.1 \text{ MPa}$$

$$\leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

6. Clamping Bolt Core Diameter (d_b)

Torque transmission by friction force balance

$$T = \frac{\pi^2}{16} \cdot \mu \cdot d \cdot \left[\frac{\pi}{4} d_b^2 \cdot \sigma_{\text{tb}}\right] \cdot n$$

$$2865 \times 10^3 = \frac{\pi^2}{16} \times 0.3 \times 75 \times \left[\frac{\pi}{4} d_b^2 \times 70\right] \times 6$$

$$2865 \times 10^3 = 5830 d_b^2$$

$$d_b^2 = 492$$

$$d_b = \textbf{22.2 mm}$$

7. Standard Bolt Selection (Table 11.1)

Standard coarse series thread lookup

$$d_{\text{c, std}} \geq d_b$$

$$= 22.2 \text{ mm}$$

$$d_c = 23.320 \text{ mm}$$

$$d = 27 \text{ mm}$$

8. Design Output

Final Design: Shaft $d = 75 \text{ mm}$, Split Muff $D = 165 \text{ mm}$, $L = 265 \text{ mm}$, 6 Bolts of M 27 Size

SECTION 2: Clamp / Split-Muff Coupling — MECHANICAL DIAGRAM

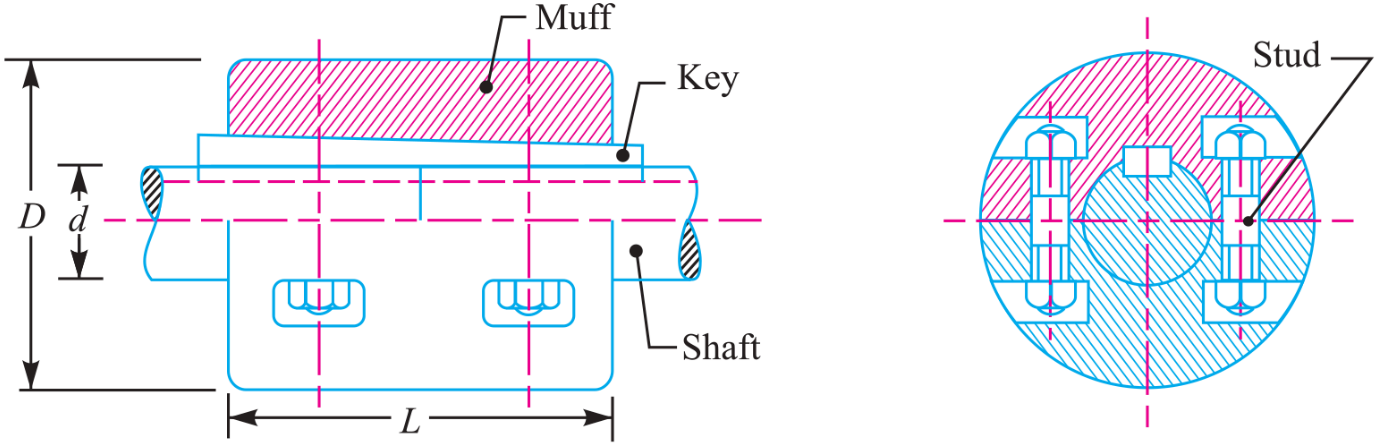

Fig. 13.11. Clamp or compression coupling.

Figure 13.2: Clamp / Split-Muff Compression Coupling Assembly & Bolt Layout

SECTION 3: Cast Iron Protective Flange Coupling Design

System Parameters

Power: $P = 15\ \text{kW} = 15000\ \text{W}$

Speed: $N = 900\ \text{rpm}$

Service Factor: $\text{K}_s = 1.35$

Shaft, Bolt & Key Shear: $\tau = 40\ \text{MPa}$

Crushing Stress: $\sigma_c = 80\ \text{MPa}$ | Cast Iron Shear: $\tau_c = 8\ \text{MPa}$

Design Protocol

Part 1: Calculate $T_{\text{max}} = 1.35 T_{\text{mean}}$ and solve shaft diameter d

Part 2: Determine hub dimensions D, L and check CI shear stress τ_c

Part 3: Verify key shearing and crushing stresses

Part 4: Calculate web thickness t_f, bolt size d_1 ($n = 3$), and outer flange D_2, t_p

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1. Mean & Maximum Design Torque (T_{max})

Maximum torque incorporating service factor

$$T_{\text{mean}} = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 15000}{2\pi \times 900}$$

$$= 159.13\ \text{N} \cdot \text{m}$$

$$T_{\text{max}} = 1.35 \times T_{\text{mean}}$$
$$= 1.35 \times 159.13$$
$$= 215 \text{ N} \cdot \text{m}$$
$$= 215 \times 10^3 \text{ N} \cdot \text{mm}$$

2. Shaft Diameter Sizing (d)

Torsional shear strength evaluation

$$d = \sqrt[3]{\frac{16 T_{\text{max}}}{\pi \tau_s}}$$
$$= \sqrt[3]{\frac{16 \times 215 \times 10^3}{\pi \times 40}}$$
$$= 30.1 \text{ mm}$$

3. Standard Shaft Diameter Selection

Select standard nominal shaft diameter

Adopt Standard Shaft Diameter $d = 35$ mm

4. Cast Iron Hub Proportions (D, L) & Shear Check (τ_c)

Hub dimensions and hollow shaft shear check

$$D = 2d = 2(35) = 70 \text{ mm}, \quad L = 1.5d = 52.5 \text{ mm}$$

$$\tau_c = \frac{16 T_{\max} D}{\pi(D^4 - d^4)}$$

$$= \frac{16 \times 215 \times 10^3 \times 70}{\pi(70^4 - 35^4)}$$

$$= 3.4 \text{ MPa}$$

$$\leq 8 \text{ MPa} \quad (\textbf{SAFE})$$

5. Key Design & Stress Verification (τ_k, σ_{ck})

Table 13.1 square key ($w = 12$ mm, $t = 12$ mm, $l = 52.5$ mm)

$$\tau_k = \frac{T_{\max}}{l \cdot w \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{215 \times 10^3}{52.5 \times 12 \times 17.5}$$

$$= 19.5 \text{ MPa}$$

$$\leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

$$\sigma_{\text{ck}} = \frac{T_{\max}}{l \cdot \left(\frac{t}{2}\right) \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{215 \times 10^3}{52.5 \times 6 \times 17.5}$$

$$= 39 \text{ MPa}$$

$$\leq 80 \text{ MPa} \quad (\textbf{SAFE})$$

6. Flange Web Thickness (t_f) & Shear Check (τ_c)

Circumferential flange web shear check

$$t_f = 0.5d = 0.5(35) = 17.5 \text{ mm}$$

$$\tau_c = \frac{2T_{\text{max}}}{\pi D^2 t_f}$$

$$= \frac{2 \times 215 \times 10^3}{\pi \times 70^2 \times 17.5}$$

$$= 1.6 \text{ MPa}$$

$$\leq 8 \text{ MPa} \quad (\textbf{SAFE})$$

7. Coupling Bolts Sizing (n, D_1, d_1)

Bolt circle geometry for $d = 35$ mm ≤ 40 mm

$$n = 3 \text{ bolts}, \quad D_1 = 3d = 3(35) = 105 \text{ mm}$$

$$T_{\text{max}} = n \cdot \left[\frac{\pi}{4} d_1^2\right] \cdot \tau_b \cdot \left(\frac{D_1}{2}\right)$$

$$215 \times 10^3 = 3 \times \left[\frac{\pi}{4} d_1^2\right] \times 40 \times \left(\frac{105}{2}\right)$$

$$d_1^2 = 43.43$$

$$d_1 = 6.6 \text{ mm} \qquad \Rightarrow \textbf{Select M 8}$$

8. Outer Flange Proportions (D_2, t_p)

Protective flange outer dimensions

$$D_2 = 4d = 4(35) = 140 \text{ mm}$$

$$t_p = 0.25d = 0.25(35) = 8.75 \text{ mm} \approx 10 \text{ mm}$$

9. Design Output

Final Design: Shaft $d = 35 \text{ mm}$, Hub $D = 70 \text{ mm}$, $L = 52.5 \text{ mm}$, Flange $t_f = 17.5 \text{ mm}$, $D_2 = 140 \text{ mm}$, 3 Bolts of M 8 Size

SECTION 3: Cast Iron Protective Flange Coupling — MECHANICAL DIAGRAM

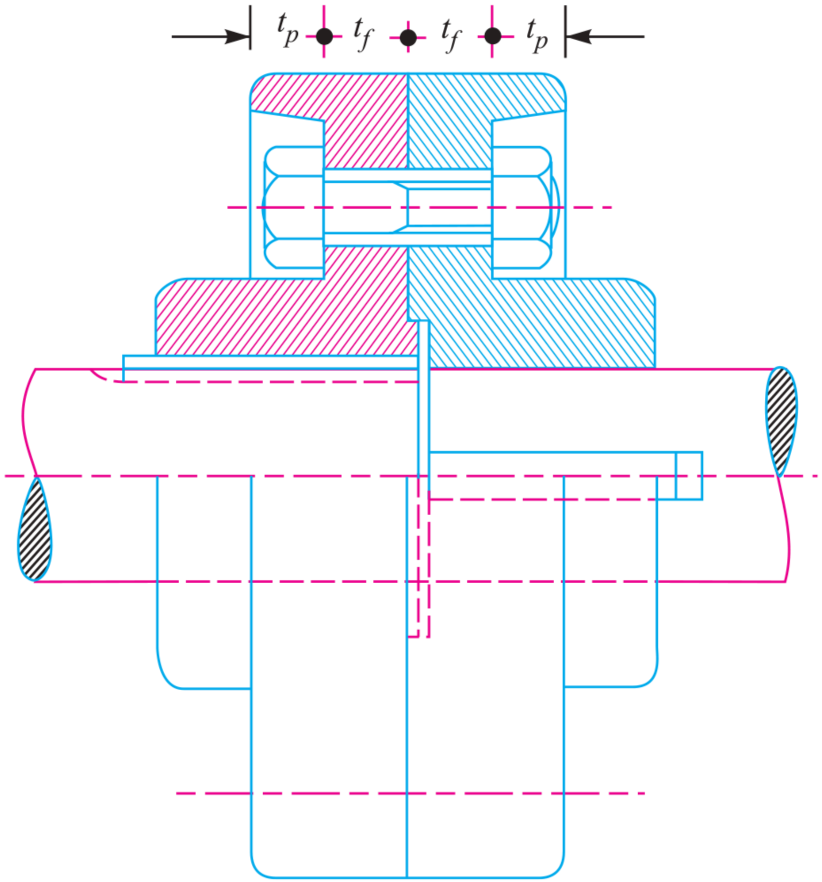

Fig. 13.13. Protective type flange coupling.

Figure 13.3: Cast Iron Protective Type Flange Coupling Mechanical Diagram

SECTION 4: Protective Flange Coupling Sizing for Peak Torque

System Parameters

Power: $P = 15\ \text{kW} = 15000\ \text{W}$

Speed: $N = 200\ \text{rpm}$

Overload: $T_{\text{max}} = 1.25 T_{\text{mean}}$

Shaft & Key Shear: $\tau = 40\ \text{MPa}$

Bolt Shear: $\tau_b = 30\ \text{MPa}$

Key Crushing: $\sigma_c = 80\ \text{MPa}$ | CI Shear: $\tau_c = 14\ \text{MPa}$

Design Protocol

Part 1: Calculate peak torque T_{max} and size shaft diameter d

Part 2: Determine hub dimensions D, L and check CI shear stress τ_c

Part 3: Verify key shearing τ_k and crushing σ_{ck}

Part 4: Calculate web thickness t_f, bolt size d_1 ($n = 4$), and outer flange D_2, t_p

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1. Peak Design Torque (T_{max})

Maximum torque calculation

$$T_{\text{mean}} = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 15000}{2\pi \times 200}$$

$$= 716\ \text{N} \cdot \text{m}$$

$$= 716 \times 10^3\ \text{N} \cdot \text{mm}$$

$$T_{\max} = 1.25 \times T_{\text{mean}}$$
$$= 1.25 \times 716 \times 10^3$$
$$= 895 \times 10^3 \ \text{N} \cdot \text{mm}$$

2. Shaft Diameter Sizing (d)

Torsional strength calculation

$$d = \sqrt[3]{\frac{16T_{\max}}{\pi\tau_s}}$$
$$= \sqrt[3]{\frac{16 \times 895 \times 10^3}{\pi \times 40}}$$
$$= 48.4 \ \text{mm}$$

3. Standard Shaft Diameter Selection

Select standard nominal shaft diameter

Adopt Standard Shaft Diameter $d = 50$ mm

4. Cast Iron Hub Proportions (D, L) & Shear Check (τ_c)

Hub dimensions and hollow shaft shear check

$$D = 2d = 2(50) = 100 \text{ mm}, \quad L = 1.5d = 75 \text{ mm}$$

$$\tau_c = \frac{16 T_{\max} D}{\pi(D^4 - d^4)}$$

$$= \frac{16 \times 895 \times 10^3 \times 100}{\pi(100^4 - 50^4)}$$

$$= 4.86 \text{ MPa}$$

$$\leq 14 \text{ MPa} \quad (\textbf{SAFE})$$

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5. Key Design & Stress Verification ($\tau_k, \sigma_{\text{ck}}$)

Table 13.1 square key ($w = 16$ mm, $t = 16$ mm, $l = 75$ mm)

$$\tau_k = \frac{T_{\max}}{l \cdot w \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{895 \times 10^3}{75 \times 16 \times 25}$$

$$= 29.8 \text{ MPa}$$

$$\leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

$$\sigma_{\text{ck}} = \frac{T_{\max}}{l \cdot \left(\frac{t}{2}\right) \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{895 \times 10^3}{75 \times 8 \times 25}$$

$$= 59.6 \text{ MPa}$$

$$\leq 80 \text{ MPa} \quad (\textbf{SAFE})$$

6. Flange Web Thickness (t_f) & Shear Check (τ_c)

Flange web shear stress check

$$t_f = 0.5d = 0.5(50) = 25 \text{ mm}$$

$$\tau_c = \frac{2T_{\max}}{\pi D^2 t_f}$$

$$= \frac{2 \times 895 \times 10^3}{\pi \times 100^2 \times 25}$$

$$= 2.5 \text{ MPa}$$

$$\leq 14 \text{ MPa} \quad (\textbf{SAFE})$$

7. Coupling Bolts Sizing (n, D_1, d_1)

Bolt circle geometry for $d = 50$ mm

$$n = 4 \text{ bolts}, \quad D_1 = 3d = 3(50) = 150 \text{ mm}$$

$$T_{\max} = n \cdot \left[\frac{\pi}{4} d_1^2\right] \cdot \tau_b \cdot \left(\frac{D_1}{2}\right)$$

$$895 \times 10^3 = 4 \times \left[\frac{\pi}{4} d_1^2\right] \times 30 \times \left(\frac{150}{2}\right)$$

$$d_1^2 = 126.6$$

$$d_1 = 11.25 \text{ mm} \qquad \Rightarrow \textbf{Select M 12}$$

8. Outer Flange Proportions (D_2, t_p)

Protective flange outer dimensions

$$D_2 = 4d = 4(50) = 200 \text{ mm}$$

$$t_p = 0.25d = 0.25(50) = 12.5 \text{ mm}$$

9. Design Output

Final Design: Shaft $d = 50\ \mathrm{mm}$, Hub $D = 100\ \mathrm{mm}$, $L = 75\ \mathrm{mm}$, Flange $t_f = 25\ \mathrm{mm}$, $D_2 = 200\ \mathrm{mm}$, 4 Bolts of M 12 Size

SECTION 4: Protective Flange Coupling — MECHANICAL DIAGRAM

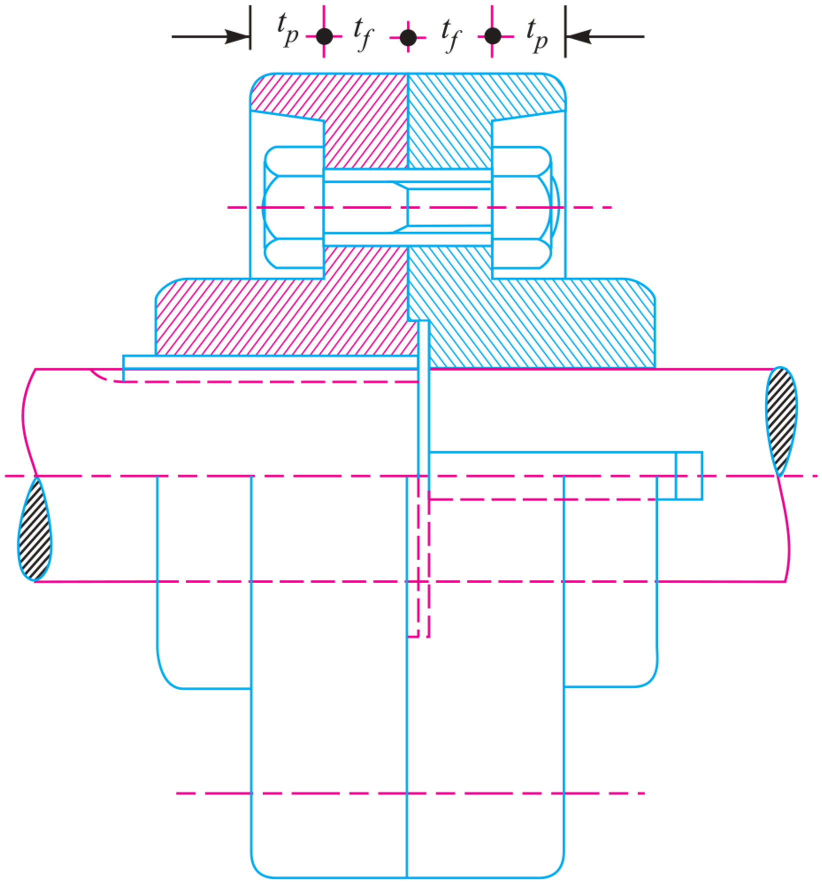

Fig. 13.13. Protective type flange coupling.

Figure 13.4: Protective Type Flange Coupling Mechanical Assembly

SECTION 5: Flange Coupling Design under Torsional Rigidity Constraint

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System Parameters

Power: $P = 90 \text{ kW} = 90000 \text{ W}$

Speed: $N = 250 \text{ rpm}$

Shaft Shear: $\tau_s = 40 \text{ MPa}$

Rigidity Limit: $\theta \leq 1^{\text{deg}} = 0.0175$ rad in length $L = 20d$

Modulus of Rigidity: $G = 84\text{kN/mm}^2$ | Bolt Shear: $\tau_b = 30 \text{ MPa}$

Design Protocol

Part 1: Calculate torque T and evaluate shaft diameter d for both strength and torsional rigidity

Part 2: Determine hub dimensions D, L and check CI shear stress τ_c

Part 3: Select key size from Table 13.1 and verify shear stress τ_k

Part 4: Calculate web thickness t_f, bolt size d_1 ($n = 4$), and outer flange D_2, t_p

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1. Transmitted Design Torque (T)

Torsional moment calculation

$$T = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 90000}{2\pi \times 250}$$

$$= 3440 \text{ N} \cdot \text{m}$$

$$= 3440 \times 10^3 \text{ N} \cdot \text{mm}$$

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2. Shaft Sizing for Strength & Rigidity (d)

Dual criterion evaluation

(a) Strength: $$d = \sqrt[3]{\frac{16T}{\pi\tau_s}}$$

$$= \sqrt[3]{\frac{16 \times 3440 \times 10^3}{\pi \times 40}}$$

$$= 76 \text{ mm}$$

(b) Rigidity: $$\frac{T}{J} = \frac{G\theta}{20d}$$

$$\frac{3440 \times 10^3}{\frac{\pi}{32}d^4} = \frac{84 \times 10^3 \times 0.0175}{20d}$$

$$d^3 = 0.476 \times 10^6$$

$$d = 78 \text{ mm}$$

3. Governing Shaft Diameter Selection

Select larger diameter for strength + rigidity

Adopt Standard Shaft Diameter $d = 80$ mm

4. Cast Iron Hub Proportions (D, L) & Shear Check (τ_c)

Hub dimensions and shear stress check

$$D = 2d = 160\ \text{mm}, \quad L = 1.5d = 120\ \text{mm}$$

$$\tau_c = \frac{16TD}{\pi(D^4 - d^4)}$$

$$= \frac{16 \times 3440 \times 10^3 \times 160}{\pi(160^4 - 80^4)}$$

$$= 4.56\ \text{MPa}$$

$$\leq 14\ \text{MPa} \quad (\textbf{SAFE})$$

5. Key Design & Stress Verification (τ_k)

Table 13.1 key ($w = 25\ \text{mm}, t = 14\ \text{mm}, l = 120\ \text{mm}$)

$$\tau_k = \frac{T}{l \cdot w \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{3440 \times 10^3}{120 \times 25 \times 40}$$

$$= 28.7\ \text{MPa}$$

$$\leq 40\ \text{MPa} \quad (\textbf{SAFE})$$

6. Flange Web Thickness (t_f) & Shear Check (τ_c)

Flange web shear check

$$t_f = 0.5d = 40 \text{ mm}$$

$$\tau_c = \frac{2T}{\pi D^2 t_f}$$

$$= \frac{2 \times 3440 \times 10^3}{\pi \times 160^2 \times 40}$$

$$= 2.14 \text{ MPa}$$

$$\leq 14 \text{ MPa} \quad (\textbf{SAFE})$$

7. Coupling Bolts Sizing (n, D_1, d_1)

Bolt PCD geometry for $d = 80$ mm

$$n = 4 \text{ bolts}, \quad D_1 = 3d = 3(80) = 240 \text{ mm}$$

$$T = n \cdot \left[\frac{\pi}{4} d_1^2\right] \cdot \tau_b \cdot \left(\frac{D_1}{2}\right)$$

$$3440 \times 10^3 = 4 \times \left[\frac{\pi}{4} d_1^2\right] \times 30 \times \left(\frac{240}{2}\right)$$

$$d_1^2 = 304$$

$$d_1 = 17.4 \text{ mm} \qquad \Rightarrow \textbf{Select M 1}$$

8. Outer Flange Proportions (D_2, t_p)

Protective outer flange dimensions

$$D_2 = 4d = 4(80) = 320 \text{ mm}$$

$$t_p = 0.25d = 0.25(80) = 20 \text{ mm}$$

9. Design Output

Final Design: Shaft $d = 80\ \mathrm{mm}$, Hub $D = 160\ \mathrm{mm}$, $L = 120\ \mathrm{mm}$, Flange $t_f = 40\ \mathrm{mm}$, $D_2 = 320\ \mathrm{mm}$, 4 Bolts of M 18 Size

SECTION 5: Unprotective Flange Coupling — MECHANICAL DIAGRAM

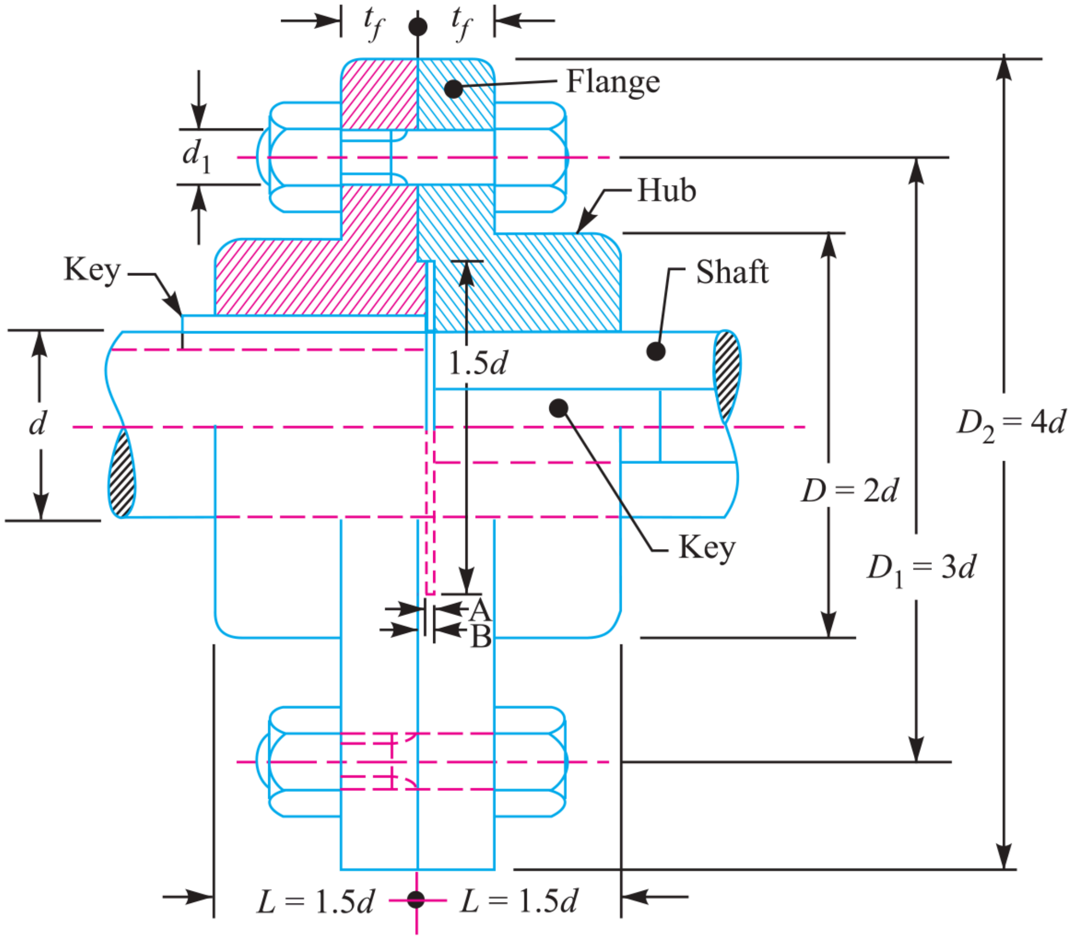

Fig. 13.12. Unprotected type flange coupling.

Figure 13.5: Unprotective Type Flange Coupling Mechanical Diagram

SECTION 6: Rigid Flange Coupling Design for Direct Torque

System Parameters

Direct Transmitted Torque: $T = 250\ \mathrm{N\cdot m} = 250\times10^3\ \mathrm{N\cdot mm}$

Bolt Count: $n = 4$

Shaft & Key Shear: $\tau = 100\ \mathrm{MPa}$

Crushing Stress: $\sigma_c = 250\ \mathrm{MPa}$

Cast Iron Shear: $\tau_c = 200\ \mathrm{MPa}$ | Bolt Shear: $\tau_b = 100\ \mathrm{MPa}$

Design Protocol

Part 1: Solve shaft diameter d for given torque T

Part 2: Determine hub dimensions D, L and check CI shear stress τ_c

Part 3: Select key dimensions from Table 13.1 and verify shear τ_k and crushing σ_{ck}

Part 4: Calculate web thickness t_f, bolt size d_1, and outer flange D_2, t_p

---

1. Shaft Diameter Sizing (d)

Torsional shear strength evaluation

$$d = \sqrt[3]{\frac{16T}{\pi\tau_s}}$$

$$= \sqrt[3]{\frac{16\times250\times10^3}{\pi\times100}}$$

$$= 23.35\ \mathrm{mm}$$

2. Standard Shaft Diameter Selection

Select standard nominal shaft diameter

Adopt Standard Shaft Diameter $d = 25$ mm

---

3. Cast Iron Hub Proportions (D, L) & Shear Check (τ_c)

Hub dimensions and shear stress check

$$D = 2d = 50 \text{ mm}, \quad L = 1.5d = 37.5 \text{ mm}$$

$$\tau_c = \frac{16TD}{\pi(D^4 - d^4)}$$

$$= \frac{16 \times 250 \times 10^3 \times 50}{\pi(50^4 - 25^4)}$$

$$= 10.86 \text{ MPa}$$

$$\leq 200 \text{ MPa} \quad (\textbf{SAFE})$$

---

4. Key Design & Stress Verification ($\tau_k, \sigma_{\text{ck}}$)

Table 13.1 key ($w = 10$ mm, $t = 8$ mm, $l = 37.5$ mm)

$$\tau_k = \frac{T}{l \cdot w \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{250 \times 10^3}{37.5 \times 10 \times 12.5}$$

$$= 53.3 \text{ MPa}$$

$$\leq 100 \text{ MPa} \quad (\textbf{SAFE})$$

$$\sigma_{\text{ck}} = \frac{T}{l \cdot \left(\frac{t}{2}\right) \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{250 \times 10^3}{37.5 \times 4 \times 12.5}$$

$$= 133.3 \text{ MPa}$$

$$\leq 250 \text{ MPa} \quad (\textbf{SAFE})$$

5. Flange Web Thickness (t_f) & Shear Check (τ_c)

Flange web shear check

$$t_f = 0.5d = 12.5 \text{ mm}$$

$$\tau_c = \frac{2T}{\pi D^2 t_f}$$

$$= \frac{2 \times 250 \times 10^3}{\pi \times 50^2 \times 12.5}$$

$$= 5.1 \text{ MPa}$$

$$\leq 200 \text{ MPa} \quad (\textbf{SAFE})$$

6. Coupling Bolts Sizing (n, D_1, d_1)

Bolt circle geometry for $d = 25$ mm

$$n = 4 \text{ bolts}, \quad D_1 = 3d = 3(25) = 75 \text{ mm}$$

$$T = n \cdot \left[\frac{\pi}{4} d_1^2\right] \cdot \tau_b \cdot \left(\frac{D_1}{2}\right)$$

$$250 \times 10^3 = 4 \times \left[\frac{\pi}{4} d_1^2\right] \times 100 \times \left(\frac{75}{2}\right)$$

$$d_1^2 = 21.22$$

$$d_1 = 4.6 \text{ mm} \qquad \Rightarrow \textbf{Select M 6 B}$$

7. Outer Flange Proportions (D_2, t_p)

Protective flange outer dimensions

$$D_2 = 4d = 4(25) = 100 \text{ mm}$$

$$t_p = 0.25d = 0.25(25) = 6.25 \text{ mm}$$

8. Design Output

Final Design: Shaft $d = 25$ mm, Hub $D = 50$ mm, $L = 37.5$ mm, Flange $t_f = 12.5$ mm, $D_2 = 100$ mm, 4 Bolts of M 6 Size

SECTION 6: Marine Flange Coupling — MECHANICAL DIAGRAM

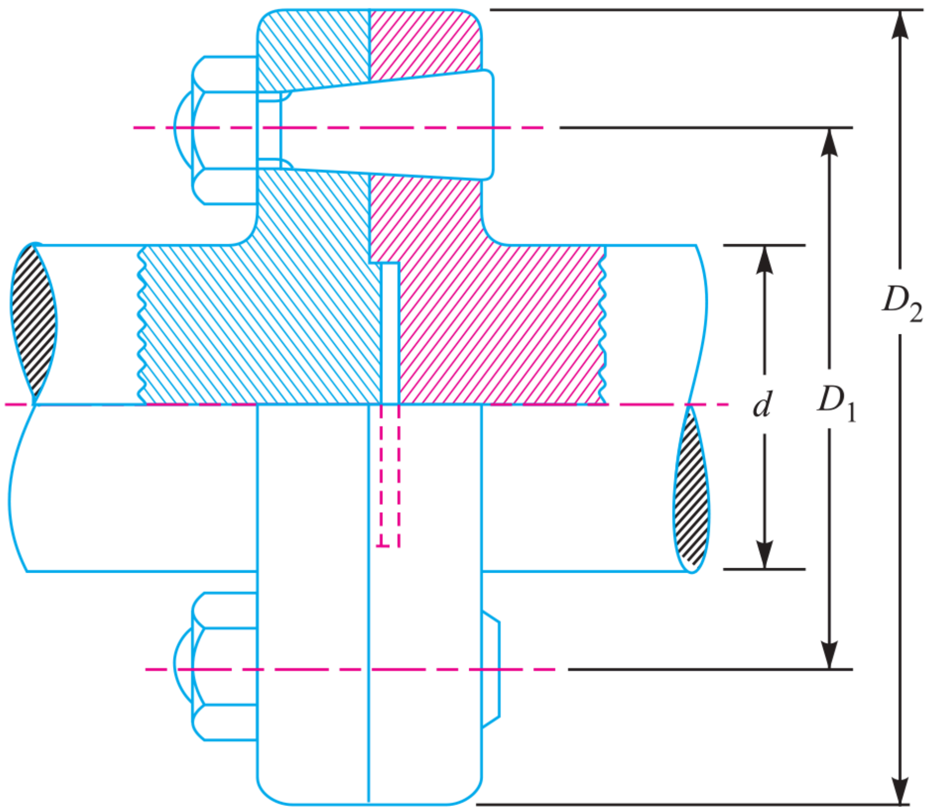

Fig. 13.14. Marine type flange coupling.

Figure 13.6: Marine Type Flange Coupling Mechanical Assembly Diagram

SECTION 7: Flanged Coupling Analysis for Fixed Shaft Diameter

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System Parameters

Fixed Shaft Diameter: $d = 35\ \text{mm}$

Transmitted Torque: $T = 800\ \text{N} \cdot \text{m} = 800 \times 10^3\ \text{N} \cdot \text{mm}$

Speed: $N = 350\ \text{rpm}$

Bolts: $n = 6$ on PCD $D_1 = 125\ \text{mm}$ | Stresses: Shaft $\tau_s = 63\ \text{MPa}$, Bolt $\tau_b = 56\ \text{MPa}$, CI $\tau_c = 10\ \text{MPa}$, Key $\tau_k = 46\ \text{MPa}$

Design Protocol

Part 1: Size bolt diameter d_1 from torque equation

Part 2: Determine web thickness t_f from hub shear

Part 3: Verify key length l against allowable shear stress τ_k and modify l if unsafe

Part 4: Match hub length $L = l$ and calculate transmitted power P

---

1. Bolt Diameter Sizing (d_1)

Torque capacity for $n = 6$ bolts on PCD $D_1 = 125\ \text{mm}$

$$T = n \cdot \left[\frac{\pi}{4} d_1^2\right] \cdot \tau_b \cdot \left(\frac{D_1}{2}\right)$$

$$800 \times 10^3 = 6 \times \left[\frac{\pi}{4} d_1^2\right] \times 56 \times \left(\frac{125}{2}\right)$$

$$d_1^2 = 48.5$$

$$d_1 = 6.96\ \text{mm}$$

$\Rightarrow$ **Select M 8 Bolts**

2. Flange Web Thickness (t_f)

Hub diameter $D = 2d = 70$ mm shear force check

$$T = \frac{\pi}{2} D^2 t_f \tau_c$$

$$800 \times 10^3 = \frac{\pi}{2}(70)^2 \times t_f \times 10$$

$$t_f = 10.4 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Flange Thickness } \; t_f = \mathbf{12 \text{ mm}}$$

3. Key Trial Length (l) & Shear Verification

Table 13.1 for $d = 35$ mm ($w = 12$ mm, $t = 8$ mm)

Trial Length $\; l = L = 1.5d = 52.5$ mm

$$\tau_k = \frac{T}{l \cdot w \cdot \left(\frac{d}{2}\right)}$$

$$= \frac{800 \times 10^3}{52.5 \times 12 \times 17.5}$$

$$= 72.5 \text{ MPa} > 46 \text{ MPa} \quad (\mathbf{U}$$

4. Recalculated Safe Key Length (l)

Solve required key length for allowable $\tau_k =$ 46 MPa

$$T = l \cdot w \cdot \tau_k \cdot \left(\frac{d}{2}\right)$$

$$800 \times 10^3 = l \times 12 \times 46 \times 17.5$$

$$l = 82.8 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Key Length } \; l = \mathbf{85 \text{ mm}}$$

5. Hub Length (L)

Set hub length equal to extended key length

Adopt Hub Length $L = l = 85$ mm

6. Transmitted Power Capacity (P)

Power transmitted at $N = 350$ rpm

$$P = \frac{2\pi NT}{60}$$

$$= \frac{2\pi \times 350 \times 800}{60}$$

$$= 29325 \text{ W}$$

$$= \mathbf{29.325 \text{ kW}}$$

7. Design Output

Final Answers: (1) M 8 Bolts, (2) $t_f = 12$ mm, (3) Key $12 \times 8 \times 85$ mm, (4) Hub $L = 85$ mm, (5) Power $P = 29.325$ kW

SECTION 8: Marine Engine Forged Flange Coupling Design

System Parameters

Engine Power: $P = 3\ \text{MW} = 3 \times 10^6\ \text{W}$

Speed: $N = 100\ \text{rpm}$

Permissible Shear Stress: Shaft & Bolts $\tau_s = \tau_b = 60\ \text{MPa}$ | Bolt Count: $n = 8$ | Pitch Circle Diameter: $D_1 = 1.6d$

Design Protocol

Part 1: Calculate design torque T and solve shaft diameter d

Part 2: Size fitted bolt diameter d_1 for $n = 8$ on PCD $D_1 = 1.6d$

Part 3: Calculate flange thickness $t_f = 0.25d$ and verify shear stress τ_c

Part 4: Calculate outer flange diameter $D_2 = D_1 + 2.5d_1$

---

1. Transmitted Design Torque (T)

High power marine engine torque calculation

$$T = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 3 \times 10^6}{2\pi \times 100}$$

$$= 286479\ \text{N} \cdot \text{m}$$

$$= 286.48 \times 10^6\ \text{N} \cdot \text{mm}$$

2. Marine Shaft Diameter Sizing (d)

Solid shaft torsional strength evaluation

$$d = \sqrt[3]{\frac{16T}{\pi \tau_s}}$$

$$= \sqrt[3]{\frac{16 \times 286.48 \times 10^6}{\pi \times 60}}$$

$$= 290 \text{ mm}$$

3. Standard Shaft Diameter Selection

Select standard nominal marine shaft diameter

Adopt Standard Shaft Diameter $d = 300$ mm

4. Coupling Bolts PCD & Shank Diameter (D_1, d_1)

Torque capacity for $n = 8$ fitted bolts on PCD $D_1 = 1.6d$

$$D_1 = 1.6d = 1.6(300) = 480 \text{ mm}$$

$$T = n \cdot \left[\frac{\pi}{4} d_1^2\right] \cdot \tau_b \cdot \left(\frac{D_1}{2}\right)$$

$$286.48 \times 10^6 = 8 \times \left[\frac{\pi}{4} d_1^2\right] \times 60 \times \left(\frac{480}{2}\right)$$

$$d_1^2 = 316.6$$

$$d_1 = 17.8 \text{ mm} \qquad \Rightarrow \textbf{Adopt Tapered B}$$

5. Forged Flange Thickness (t_f) & Shear Check (τ_c)

Flange web shear stress check at junction

$$t_f = 0.25d = 0.25(300) = 75 \text{ mm}$$

$$\tau_c = \frac{2T}{\pi d^2 t_f}$$

$$= \frac{2 \times 286.48 \times 10^6}{\pi \times 300^2 \times 75}$$

$$= 27 \text{ MPa}$$

$$\leq 60 \text{ MPa} \quad (\textbf{SAFE})$$

6. Outer Flange Diameter (D_2)

Overall diameter of marine flange

$$D_2 = D_1 + 2.5d_1$$

$$= 480 + 2.5(60)$$

$$= 630 \text{ mm}$$

7. Design Output

Final Answers: (1) Shaft $d = 300$ mm, (2) 8 Tapered Bolts $d_1 = 60$ mm, (3) Flange $t_f = 75$ mm, (4) Flange $D_2 = 630$ mm

SECTION 8: Marine Engine Forged Flange Coupling — MECHANICAL DIAGRAM

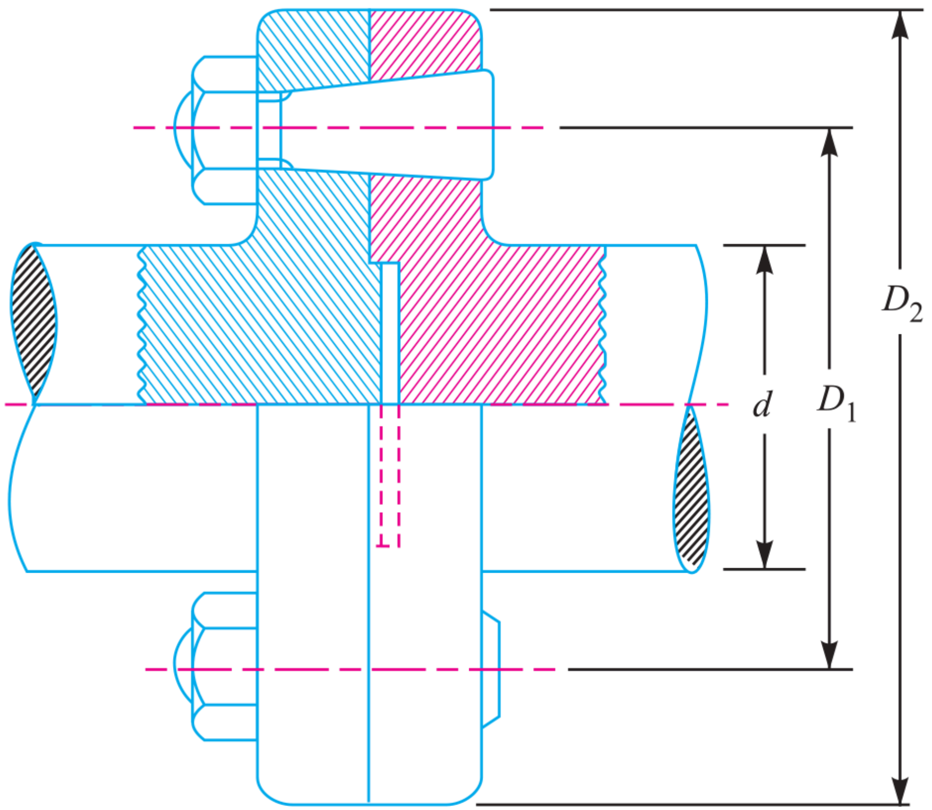

Fig. 13.14. Marine type flange coupling.

Figure 13.6: Marine Engine Forged Flange Coupling Mechanical Assembly Diagram

SECTION 9: Bushed-Pin Flexible Coupling Design

System Parameters

Power: $P = 32\ \text{kW} = 32000\ \text{W}$

Speed: $N = 960\ \text{rpm}$

Overload Capacity: $1.20 T_{\text{mean}}$

Shaft, Key & Pin Shear: $\tau = 40\ \text{MPa}$

Key Crushing: $\sigma_c = 80\ \text{MPa}$

CI Shear: $\tau_c = 15\ \text{MPa}$ | Rubber Bush Bearing Pressure: $p_b = 0.8\text{N/mm}^2$

Design Protocol

Part 1: Calculate peak torque T_{max} and size shaft diameter d

Part 2: Layout flexible pins ($n = 6, d_1 = 20\ \text{mm}$), bush $d_2 = 40\ \text{mm}$, PCD $D_1 = 132\ \text{mm}$, and solve bush length l

Part 3: Verify pin direct shear, bending stress, and principal stresses

Part 4: Determine hub dimensions D, L, key w, t, l, and flange web t_f

---

1. Peak Design Torque (T_{max})

Torque calculation incorporating 20% overload

$$T_{\text{mean}} = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 32000}{2\pi \times 960}$$

$$= 318.3\ \text{N} \cdot \text{m}$$

$$T_{\text{max}} = 1.20 \times T_{\text{mean}}$$
$$= 1.20 \times 318.3$$
$$= 382 \text{ N} \cdot \text{m}$$
$$= 382 \times 10^3 \text{ N} \cdot \text{mm}$$

2. Shaft Diameter Sizing (d)

Torsional strength evaluation

$$d = \sqrt[3]{\frac{16T_{\text{max}}}{\pi\tau_s}}$$
$$= \sqrt[3]{\frac{16 \times 382 \times 10^3}{\pi \times 40}}$$
$$= 36.5 \text{ mm}$$

3. Standard Shaft Diameter Selection

Select standard nominal shaft diameter

Adopt Standard Shaft Diameter $d = 40$ mm

4. Pin & Rubber Bush Proportions (n, d_1, d_2, D_1)

Flexible pin geometry

$$n = 6 \text{ pins}, \quad d_1 = 20 \text{ mm} \quad \text{(Pin Shank)}$$
$$d_2 = 40 \text{ mm} \quad \text{(Rubber Bush Outer Diameter)}$$
$$D_1 = 2d + d_2 + 2(6) = 2(40) + 40 + 12 = 132 \text{ mm}$$

5. Bush Length (l) & Bearing Load (W)

Bearing pressure and torque moment equilibrium

$$W = p_b \cdot d_2 \cdot l = 0.8 \times 40 \times l = 32l \text{ N}$$

$$T_{\max} = W \cdot n \cdot \left(\frac{D_1}{2}\right)$$

$$382 \times 10^3 = (32l) \times 6 \times \left(\frac{132}{2}\right)$$

$$382 \times 10^3 = 12672l$$

$$l = 30.1 \text{ mm} \qquad \Rightarrow \textbf{Adopt Bush Len}$$

$$W = 32(32) = \mathbf{1024\ N}$$

6. Flexible Pin Direct Shear Stress (τ)

Direct shear force on pin shank

$$\tau = \frac{W}{\frac{\pi}{4} d_1^2}$$

$$= \frac{1024}{\frac{\pi}{4}(20)^2}$$

$$= 3.26 \text{N/mm}^2$$

7. Flexible Pin Bending Moment (M) & Bending Stress (σ_b)

Bending moment and section modulus

$$M = W\left(\frac{l}{2} + 5\right)$$

$$= 1024(16 + 5)$$

$$= 21504 \text{ N} \cdot \text{mm}$$

$$Z = \frac{\pi}{32} d_1^3$$

$$= \frac{\pi}{32} (20)^3$$

$$= 785.5 \text{mm}^3$$

$$\sigma_b = \frac{M}{Z}$$

$$= \frac{21504}{785.5}$$

$$= 27.4 \text{N/mm}^2$$

8. Flexible Pin Maximum Principal Stress ($\sigma_{\max}$)

Maximum principal stress theory check

$$\sigma_{\max} = \frac{1}{2}\left[\sigma_b + \sqrt{\sigma_b^2 + 4\tau^2}\right]$$

$$= \frac{1}{2}\left[27.4 + \sqrt{(27.4)^2 + 4(3.26)^2}\right]$$

$$= 27.8 \text{ MPa}$$

$$\leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

9. Flexible Pin Maximum Shear Stress ($\tau_{\max}$)

Maximum shear stress theory check

$$\tau_{\max} = \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$$
$$= \frac{1}{2}\sqrt{(27.4)^2 + 4(3.26)^2}$$
$$= 14.1 \text{ MPa}$$
$$\leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

10. Cast Iron Hub Proportions (D, L) & Shear Check (τ_c)

Hub dimensions and shear stress check

$$D = 2d = 80 \text{ mm}, \quad L = 1.5d = 60 \text{ mm}$$

$$\tau_c = \frac{16 T_{\max} D}{\pi(D^4 - d^4)}$$
$$= \frac{16 \times 382 \times 10^3 \times 80}{\pi(80^4 - 40^4)}$$
$$= 4.05 \text{ MPa}$$
$$\leq 15 \text{ MPa} \quad (\textbf{SAFE})$$

11. Key Design & Stress Verification (τ_k, σ_{ck})

Table 13.1 square key ($w = 14$ mm, $t = 14$ mm, $l = 60$ mm)

$$\tau_k = \frac{T_{\max}}{l \cdot w \cdot \left(\frac{d}{2}\right)} = \frac{382 \times 10^3}{60 \times 14 \times 20} = 22.74 \text{ MPa} \leq 40 \text{ MPa} \quad (\textbf{SAFE})$$

$$\sigma_{\text{ck}} = \frac{T_{\max}}{l \cdot \left(\frac{t}{2}\right) \cdot \left(\frac{d}{2}\right)} = \frac{382 \times 10^3}{60 \times 7 \times 20} = 45.48 \text{ MPa} \leq 80 \text{ MPa} \quad (\textbf{SAFE})$$

---

12. Flange Web Thickness (t_f) & Shear Check (τ_c)

Flange web shear stress check

$$t_f = 0.5d = 20 \text{ mm}$$

$$\tau_c = \frac{2T_{\max}}{\pi D^2 t_f} = \frac{2 \times 382 \times 10^3}{\pi \times 80^2 \times 20} = 1.9 \text{ MPa} \leq 15 \text{ MPa} \quad (\textbf{SAFE})$$

13. Design Output

Final Design: Shaft $d = 40 \text{ mm}$, 6 Flexible Pins $d_1 = 20 \text{ mm}$, Rubber Bush $d_2 = 40 \text{ mm}, l = 32 \text{ mm}$, Hub $D = 80 \text{ mm}, L = 60 \text{ mm}$, Flange $t_f = 20 \text{ mm}$

SECTION 8: Bushed-Pin Flexible Coupling — MECHANICAL DIAGRAM

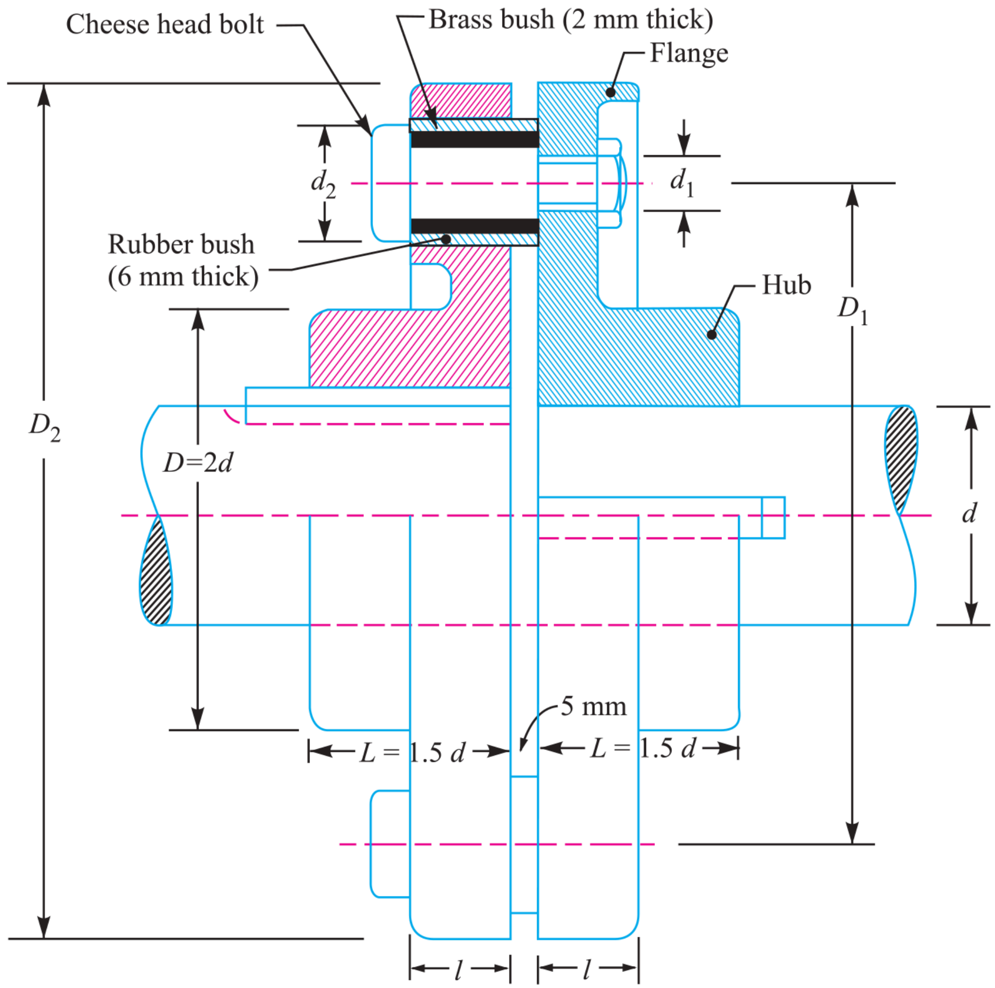

Fig. 13.15. Bushed-pin flexible coupling.

Figure 13.7: Bushed-Pin Flexible Coupling Mechanical Assembly & Rubber Bush Details